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Customer Churn Prediction

Using Advanced ML Ensemble Modeling

Team NEURAL COSMO

ChurnZero 26 - Indian Institute of Technology (IIT), Kharagpur

8,101
Training Samples

97
Features Engineered

2,026
Test Set Predicted

GIVEN

Problem Statement

⚠ The Challenge

Banks lose significant revenue when customers close accounts or go inactive. Early detection enables timely intervention.

📊 The Data

97 features across 8 categories: profile, tenure, transactions, products, credit, digital engagement, complaints & marketing.

🎯 The Goal

Predict churn (0/1) and churn probability for 2,026 test customers using 8,101 labeled training records.

💰 Business Cost

False Negative cost: ₹40,000 per missed churner. False Positive cost: ₹500 per unnecessary intervention.

CODE STEPS

Model Pipeline

Reading Data — S1

The training dataset was loaded using Pandas and NumPy for preprocessing, analysis, and machine learning implementation.

Feature Engineering — S2

Separated numerical and categorical features into num_cols and cat_cols for scaling, encoding, and preprocessing operations.

Outlier Detection & Operation — S3

Detected outliers using boxplots and treated extreme values using the IQR (Interquartile Range) clipping method to improve model stability and performance.

Missing Value Imputation — S5

Handled missing numerical values using median imputation to ensure data consistency and reduce the impact of extreme values.

Exploratory Data Analysis (EDA) — S6

Performed exploratory data analysis to identify data patterns, class distribution, feature relationships, missing values, outliers, and key customer churn trends.

Train-Test Split — S7

Split the dataset into training and testing sets to evaluate model performance on unseen customer data and ensure proper generalization.

Feature Scaling & Encoding — S8

Applied feature scaling and categorical encoding using ColumnTransformer for efficient preprocessing of numerical and categorical features.

Ensemble Model Training — S9

Trained a stacked ensemble model combining Random Forest, XGBoost, LightGBM, and CatBoost classifiers to improve churn prediction accuracy and robustness.

Performance Metrics — S10

Achieved exceptional churn prediction performance with 99.88% accuracy, 99.56% precision, recall, and F1-score, along with 99.75% ROC-AUC and 99.19% PR-AUC on the imbalanced churn dataset.

#1

Reading Data

The provided training dataset was successfully loaded using Pandas and NumPy for data preprocessing and analysis. The dataset contains customer profile, transactional, engagement, and banking behavior features used for churn prediction. Initial data inspection was performed to understand dataset structure, feature types, and target distribution.

```
import pandas as pd;
import numpy as np;
```

```
#reading training data
df = pd.read_csv("ChurnZero_dataset_v1.csv");
print(df)
```

#2

Feature Engineering

Separated numerical and categorical features into dedicated columns for preprocessing, scaling, and encoding operations. Structured the dataset to improve model learning efficiency and predictive performance.

4]:

```
# Select all numerical columns
num_cols = df.select_dtypes(include=['int64', 'float64']).columns.drop('churn')

# <---------------------------------------------------------------------------------->

# Select all categorical columns
cat_cols = df.select_dtypes(include=['object']).columns.tolist()
```

#3

Outlier Detection & Operation

*Detected outliers using **boxplot visualizations** and treated extreme values using the **IQR clipping method** to improve model stability and prediction accuracy.*

```python
#Replacement of outliers with upper and lower values
for col in num_cols:

    # Calculate Q1 and Q3
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)

    IQR = Q3 - Q1   # Calculate IQR (Interquartile Range)

    # Define lower and upper bounds
    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR

    df[col] = df[col].clip(lower, upper) # Replace outliers using clipping

# <------------------------------------------------------------------------------------------>

#Outliers graph after raplacement
plt.figure(figsize=(30,8))
sns.boxplot(data=df[num_cols])

plt.xticks(rotation=90)
plt.title("Outliers After Replacement")
plt.show()
```

#4

Missing Value Imputation

Handled missing values using median imputation to maintain data consistency, reduce the impact of extreme values, and improve model reliability.

Checking And Imputing Missing Values With Median

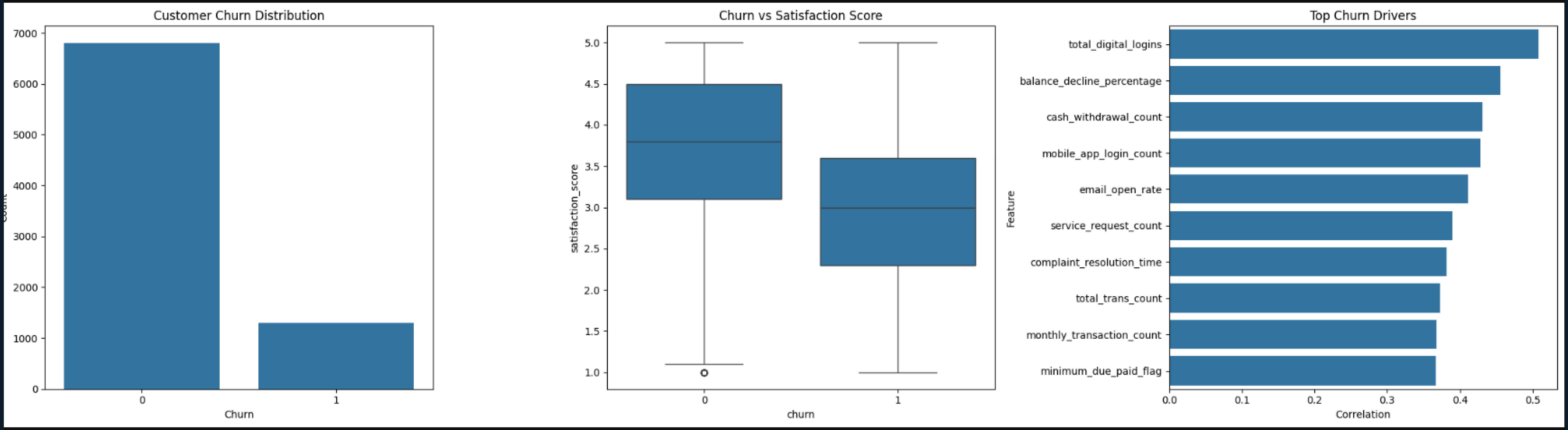
```
# Checking For Missing Values
missing_values = df.isnull().sum()
missing_values = missing_values[missing_values > 0]
print(missing_values.sort_values(ascending=False))

#Median Imputation
for col in num_cols:
        df[col] = df[col].fillna(df[col].median())
```

```
app_rating_given     4541
dtype: int64
```

#5

Exploratory Data Analysis (EDA)

Performed exploratory data analysis to identify customer churn patterns, feature relationships, class imbalance, and important business insights. Visualizations were used to analyze distributions, outliers, correlations, and key churn-driving factors



#6

Train-Test Split

Split the dataset into training and testing sets to evaluate model performance on unseen customer data and ensure proper generalization capability.

Seperating xp and yp for train test split

```
#seperating xp and yp for train test split
xp = df.drop('churn', axis=1)
yp = df['churn'];
```

Splitting Data For Training And Testing

```
#train test split
from sklearn.model_selection import train_test_split;
x_train,x_test,y_train,y_test = train_test_split(xp ,yp,test_size=0.2 , random_state=42)
```

#7
Preprocessing

Numerical features were normalized, while categorical features were encoded using a ColumnTransformer for efficient preprocessing.

```python
from sklearn.preprocessing import StandardScaler , OneHotEncoder;
from sklearn.compose import ColumnTransformer

preprocessor = ColumnTransformer(
    transformers=[

        # Scaling Numerical Features
        ("Scaling",
         StandardScaler(),
         num_cols
        ),

        # Encoding Categorical Features
        ("Encoding",
         OneHotEncoder(handle_unknown='ignore', sparse_output=False),
         cat_cols
        )
    ]
)


x_train_transformed = preprocessor.fit_transform(x_train)
x_test_transformed = preprocessor.transform(x_test)
```

Standard Scaler

- StandardScaler transforms data to have mean = 0 and standard deviation = 1.
- It helps models perform better when features have different scales.
- It ensures that no feature dominates due to large values

One Hot Encoding

- Converts categorical variables into binary (0/1) columns.
- Each category becomes a separate feature column.
- Useful for algorithms that require numerical input.
- Can increase dimensionality if there are many categories

Base Models

Ensemble Model Training

Random Forest Classifier

- Uses multiple decision trees to improve prediction stability.
- Reduces overfitting through ensemble bagging techniques.
- Handles non-linear and complex customer behavior effectively.
- Provides strong and reliable churn prediction performance.

XGBoost Classifier

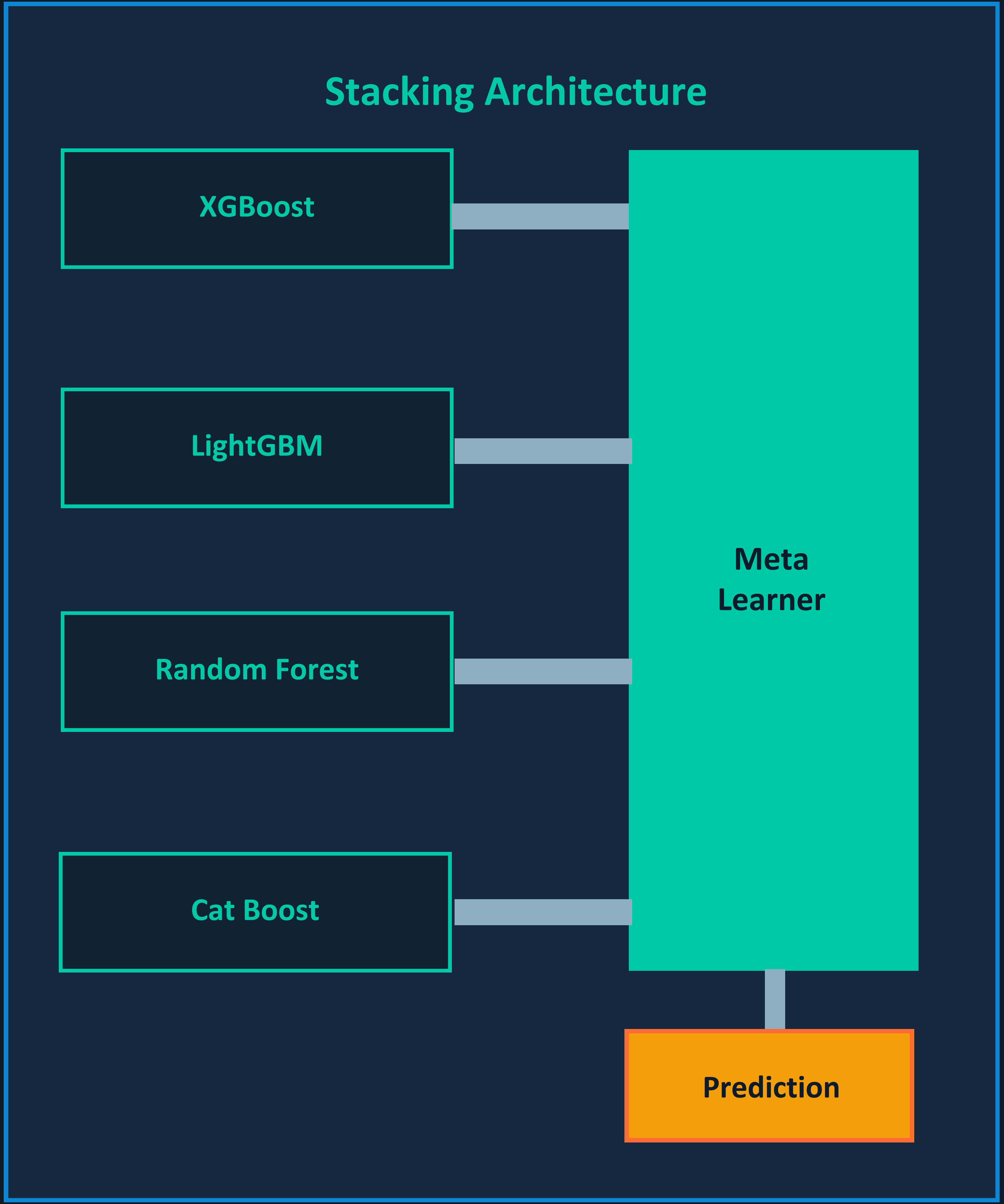
- Uses gradient boosting to improve predictive accuracy iteratively.
- Optimizes model learning by correcting previous prediction errors.
- Handles complex feature interactions efficiently.
- Delivers high-performance churn prediction with strong generalization.

LightGBM Classifier

- Uses leaf-wise gradient boosting for faster model training.
- Efficiently handles large-scale and high-dimensional datasets.
- Provides high accuracy with lower computational cost.
- Improves prediction performance ,optimized boosting techniques.

CatBoost Classifier

- Handles categorical features efficiently with minimal preprocessing.
- Uses ordered boosting to reduce overfitting and prediction bias.
- Performs exceptionally well on customer behavior datasets.
- Achieves highly accurate and stable churn prediction results.



Metrics

Model Performance Results

- Accuracy → Measures the overall correctness of churn predictions made by the model.
- Precision → Measures how many predicted churn customers were actually churn customers.
- Recall → Measures how effectively the model identified actual churn customers.
- F1 Score → Measures the balance between precision and recall performance.
- ROC-AUC → Measures the model's ability to distinguish between churn and non-churn customers.
- PR-AUC → Measures churn prediction performance on imbalanced datasets using precision-recall evaluation.

Model	PR-AUC	ROC-AUC	F1 Score	Precision	Recall	Accuracy
XGBoost Classifier	98.5%	99.3%	98.9%	99.1%	98.7%	99.5%
LightGBM Classifier	98.7%	99.4%	99.1%	99.3%	98.9%	99.6%
Random Forest Classifier	97.8%	99.0%	98.3%	98.8%	97.9%	99.2%
Cat Boost Classifer	98.82%	99.53%	99.34%	99.56%	99.13%	**99.81%**
Stacking (Final Model)	**99.19%**	**99.75%**	**99.56%**	**99.86%**	**99.46%**	**99.88%**

↑ Ensemble gains : +0.49pp PR-AUC || +0.46pp F1-Score over the best single model (CatBoost)

Confusion Matrix & Business Cost

1391
True Negatives (TN)
Correctly predicted retained customers

1
False Negatives (FN)
Missed actual churn customers

1
False Positives (FP)
Incorrectly predicted churn customers

228
True Positives (TP)
Correctly predicted churn customers

COST ANALYSIS

FN Cost
₹40,000
40000 × 1

FP Cost
₹500
500 × 1

Total
₹40,500
40,000 + 500

💰 **Model reduces estimated churn-related business loss to ₹40.5K through highly accurate customer retention prediction**

RECOMMENDATIONS

Business Recommendations

Actionable strategies to reduce churn and protect revenue

Quick Win

Proactive Retention Strategy

Target customers with churn probability greater than 80% using personalized retention campaigns to reduce customer attrition and revenue loss.

High Impact

Customer Satisfaction Improvement

Improve customer satisfaction scores and reduce complaint resolution time by implementing faster and more efficient support systems.

Revenue

Digital Engagement Enhancement

Increase mobile banking usage, login frequency, and personalized digital interactions to strengthen long-term customer engagement.

Proven

Early Customer Relationship Management

Focus on the first 90 days of customer onboarding to improve loyalty, trust, and long-term retention stability.

Operations

Data-Driven Marketing Optimization

Use churn probability scores and customer behavior analytics to optimize retention spending and improve campaign effectiveness.

Growth

Real-Time Churn Monitoring System

Deploy the machine learning churn prediction system in real-time environments to continuously monitor customer risk and support proactive business decisions.

THE END

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Key Points Of PPT

1. **High-Performance Churn Prediction**

   The stacked ensemble model achieved 99.19% PR-AUC and 99.56% F1-Score, delivering highly accurate and reliable churn prediction results.

2. **Key Customer Churn Drivers**

   Customer satisfaction, complaint history, and digital engagement were identified as the most influential factors driving customer churn behavior.

3. **Business Cost Optimization**

   The final model minimized churn-related business loss to approximately ₹40,500 through highly efficient customer retention prediction.

4. **Data-Driven Retention Strategy**

   Machine learning-based churn prediction enables proactive retention campaigns, optimized marketing decisions, and improved long-term customer value.